

Number Manipulation Functions

Sample Table Structure

```
CREATE TABLE employee_salary (  
emp_id INT PRIMARY KEY,  
emp_name VARCHAR(50),  
base_salary DECIMAL(10,2),  
bonus DECIMAL(10,2),  
tax_percent DECIMAL(5,2),  
experience_years INT  
);
```

Insert Sample Data

```
INSERT INTO employee_salary VALUES  
(1, 'Karthik', 75000.75, 5000.50, 10.00, 6),  
(2, 'Veena', 65000.40, 4000.25, 8.50, 4),  
(3, 'Ravi', 85000.90, 6000.75, 12.00, 8),  
(4, 'Anil', 70000.10, NULL, 9.00, 5),  
(5, 'Suresh', 60000.55, 3000.30, 7.50, 3);
```

NUMERIC FUNCTIONS IN MYSQL

1. ABS – Absolute Value

```
SELECT ABS(-100);
```

2. ROUND – Round to Decimals

```
SELECT ROUND(base_salary, 0) FROM employee_salary;
```

```
SELECT ROUND(base_salary, 2) FROM employee_salary;
```

3. CEILING / CEIL – Round Up

SELECT CEIL(base_salary) FROM employee_salary;

4. FLOOR – Round Down

SELECT FLOOR(base_salary) FROM employee_salary;

5. TRUNCATE – Cut Decimals (No Rounding)

SELECT TRUNCATE(base_salary, 1) FROM employee_salary;

6. MOD – Remainder

SELECT MOD(experience_years, 2) FROM employee_salary;

7. POWER / POW – Exponent

SELECT POWER(2, 3);

SELECT POW(5, 2);

8. SQRT – Square Root

SELECT SQRT(64);

9. SIGN – Sign of Number

SELECT SIGN(base_salary) FROM employee_salary;

10. RAND – Random Number

SELECT RAND();

11. FORMAT – Format Number as String

SELECT FORMAT(base_salary, 2) FROM employee_salary;

Note: FORMAT() returns a string, not a numeric value.

12. GREATEST – Maximum Value

SELECT emp_name,

GREATEST(base_salary, IFNULL(bonus, 0))

FROM employee_salary;

13. LEAST – Minimum Value

```
SELECT emp_name,  
LEAST(base_salary, IFNULL(bonus, 0))  
FROM employee_salary;
```